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End Semester Examination 2025

Name of the Course: B.Tech

Semester: 3rdName of the Paper – Electrical Machines I: **TEE 302**

Time: 3HoursMaximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b, and c in each main question
- (iii) Total marks in each main question is twenty.
- (iv) Each question carries 20 marks

Q1	(20 marks)	
(a)	Derive the emf equation of a single-phase transformer and explain the significance of each term in it. Include the role of frequency, flux, and turns ratio	CO1,CO2
(b)	Derive the condition for maximum efficiency of a single-phase transformer. Explain how efficiency varies with load.	
(c)	A 50 kVA, 2000/200 V, 50 Hz single-phase transformer has an open-circuit (no-load) input of 350 W at 2000 V and draws 1.2 A. Under short-circuit test (with secondary shorted) it requires 80 V to circulate full-load current of 25 A and the input is 620 W. Determine: i) the equivalent resistance and leakage reactance referred to the primary side, ii) the approximate efficiency at full load at 0.8 lagging power-factor, neglecting magnetizing current.	
Q2	(20 marks)	
(a)	Discuss the voltage regulation of a transformer. Derive the expression for percentage voltage regulation and explain its variation with load power factor.	CO3,CO4
(b)	Explain the different types of three-phase transformer connections (Δ - Δ , Δ -Y, Y- Δ , Y-Y). Discuss their advantages, disadvantages, and applications.	
(c)	Compare the performance of a bank of three single-phase transformers with a single three-phase transformer.	
Q3	(20 marks)	
(a)	Discuss the concept of energy balance in electromechanical systems. Starting from the basic law of conservation of energy, derive the relation between electrical input power, mechanical output power, and field energy.	CO5,CO1
(b)	Describe in detail the torque production mechanism in rotating electrical machines.	

	Explain how energy conversion takes place between electrical and mechanical domains through the magnetic field.	
(c)	For a system of multiply excited magnetic system various inductances are; $L_{11}=(4+\cos 2\theta) \times 10^{-3} \text{H}$, $L_{12}=0.25 \cos \theta \text{ H}$, $L_{22}=(10+5 \cos 2\theta) \text{H}$ Find the torque developed if $i_1=2 \text{A}$ and $i_2=0.02 \text{A}$	
Q4	(20 marks)	
(a)	A 250 V DC shunt motor has $R_a = 0.2 \Omega$. When taking an armature current of 50 A, it runs at 1000 rpm. Calculate the speed when the armature current is 30 A, assuming the flux is reduced by 5 % due to armature reaction.	CO3,CO6
(b)	Explain the starting of DC motors. Why is a starter necessary? Describe the working of a three-point or four-point starter with a neat diagram.	
(c)	Discuss the difference between a DC generator and a DC motor on the basis of construction, operation, and characteristics. Explain how the same machine can operate as either.	
Q5	(20 marks)	
(a)	Draw and explain the torque equation of a DC motor. Show how torque depends on flux and armature current.	CO2
(b)	A 220 V DC series motor draws 30 A at 800 rpm. The total resistance of the armature and series field is 0.6Ω . Find the speed when the current is 20 A, assuming the flux is proportional to the armature current.	
(c)	Describe the constructional details of a DC generator with neat and labeled diagrams of all main parts. Explain the function of each part briefly.	